

IMPACT CRATERS

Meteorite impact craters on Venus range in size from 4 miles (7 km) to 168 miles (270 km) across. The largest are multiple-ringed, those of intermediate size have central peaks, and the smaller ones have smooth floors. The smallest of all—simple, bowl-like craters that are common on the Moon and Mars—are scarce on Venus, because the thick atmosphere filters out the small asteroids that would create them.

Venusian craters are young and in many cases in pristine condition. The last volcanic resurfacing of Venus could have occurred as recently as 500 million years ago, so its craters must have mostly formed since then, and there has been little geological activity or weathering to affect them. Individual craters on Venus are named after women of note or are given female first names.

CUNITZ CRATER

This typical impact crater has a dark basin 30 miles (48 km) wide, with mountainous central peaks and a bright ejecta blanket around it.

ISHTAR TERRA

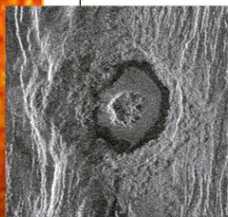
Wanda Crater



TYPE Central-peak crater
AGE Under 500 million years
DIAMETER 13.4 miles (21.6 km)

Wanda Crater is in the northern part of the Akna Montes mountain range. It was mapped first in 1984, by the Venera 15 and 16 missions, and Magellan studied it again a few years later. The crater has a large, rugged peak in the center of its smooth,

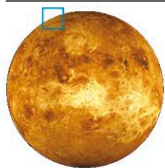
lava-flooded floor. About one-third of all Venusian craters have such peaks. Material from the mountain ridges seems to have collapsed into the crater's western edge.



CENTRAL PEAK

ISHTAR TERRA

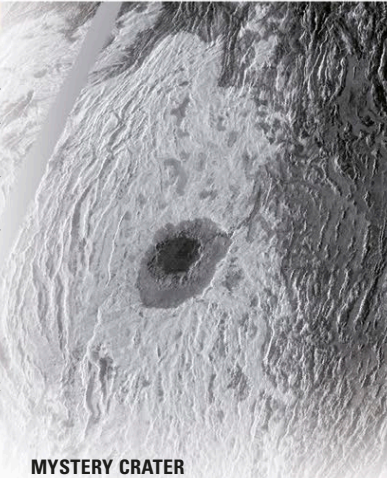
Cleopatra Crater



TYPE Double-ring crater
AGE Under 500 million years
DIAMETER 65 miles (105 km)

Cleopatra Crater is named after the legendary Egyptian queen. It is located on Maxwell Montes, Venus's highest mountain range and stands out as a smooth, eyelike feature against the rough mountainous terrain.

Cleopatra was imaged by the Venera 15 and 16 spacecraft and the Arecibo radio telescope in the mid-1980s. It was one of several circular features that resembled both an impact crater and a volcanic feature. The data of the time revealed a feature of apparently great depth without the rim deposits typical of impact craters. As a result, Cleopatra was classified as a volcanic caldera. However, high-resolution



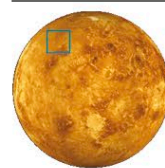
MYSTERY CRATER

The dark inner basin, the rim, and the surrounding ejecta revealed in this Magellan image from 1990 convinced astronomers that Cleopatra is an impact crater.

images from Magellan revealed an inner basin and rough ejecta deposits, providing conclusive proof that Cleopatra is an impact crater.

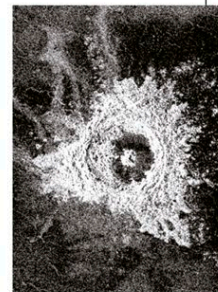
GUINEVERE PLANITIA

Jeanne Crater



TYPE Central-peak crater
AGE Under 500 million years
DIAMETER 12 miles (19.4 km)

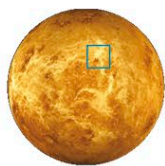
An asteroid traveling from the southwest smashed obliquely into the Guinevere Planitia and created Jeanne Crater. Ejecta pushed out of the impact basin produced a distinctive triangular shape. Lobes formed to the northwest of the crater as molten material produced by the impact flowed downhill.



TRIANGULAR EJECTA

BETA REGIO

Balch Crater



TYPE Central-peak crater
AGE Under 500 million years
DIAMETER 25 miles (40 km)

Most impact craters on Venus have remained unchanged since they were formed and have sharply defined rims. However, a relatively small number have been modified by volcanic eruptions and other kinds of tectonic activity. Balch Crater is one of these. Its circular form was split in two as the land was wrenched apart during the formation of a deep rift valley. The rift, which is up to 12.4 miles (20 km) wide, created a division that runs from north to south through the crater's center. The western half of the crater remains intact, but most of the eastern half was destroyed. A central peak and an ejecta blanket are visible in the western half. The crater was initially called Somerville but is now named after American economist and Nobel laureate Emily Balch.

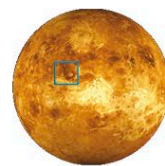


HALF CRATER

A rift valley separates most of the Balch Crater (left) from its smaller eastern part, just visible on the opposite side of the fault. The original central peak is the bright patch in the crater's western part.

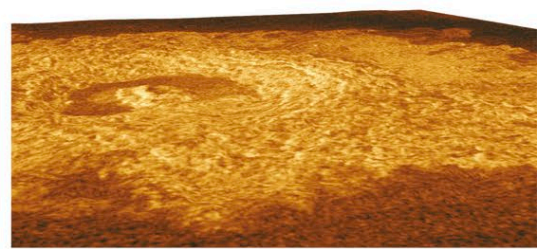
APHRODITE TERRA

Riley Crater



TYPE Central-peak crater
AGE Under 500 million years
DIAMETER 16 miles (25 km)

Riley Crater, named after 19th-century botanist Margaretta Riley, is one of the few Venusian craters to have been precisely measured. Comparison of images from different angles shows that the 16-mile (25-km) wide crater's floor lies 1,880 ft (580 m) below the surrounding plain, the rim is 2,009 ft (620 m) above it, and the peak is 1,737 ft (536 m) high.



OBLIQUE VIEW OF RILEY CRATER